SS2 PHYSICS LESSON NOTES (THIRD TERM)
WEEK 8:ENERGY AND SOCIETY

1. MEANING AND FORMS OF ENERGY

1.1 Definition of Energy

Energy is the capacity to do work or cause change. It is a fundamental physical quantity and is required in all activities of life, from motion and growth to production and communication.

Energy exists in various forms and can be transformed from one form to another.

1.2 Forms of Energy

S/N	Form of Energy	Description	Formula
1	**Kinetic Energy**	Energy possessed by a body due to its motion	$KE = \frac{1}{2}mv^2$ (where m = mass, v = velocity)
2	**Potential Energy**	Energy stored in a body due to its position or configuration	$PE = mgh$ (where m = mass, g = gravity, h = height)

S/N	Form of Energy	Description	Formula
3	**Thermal Energy**	The internal energy of an object due to the random motion of its particles; related to temperature	—
4	**Electrical Energy**	Energy caused by the movement of electric charges (electrons); powers electrical devices and systems	—
5	**Chemical Energy**	Energy stored in chemical bonds of substances; released during chemical reactions such as combustion or metabolism	—
6	**Nuclear Energy**	Energy stored in the nucleus of atoms; released during nuclear fission or fusion	$E = \Delta m \cdot c^2$
7	**Sound Energy**	Energy carried by sound waves, resulting from vibrations of particles	—
8	**Radiant (Light) Energy**	Energy carried by electromagnetic waves, especially visible light, X-rays, UV rays, etc.	—

2. MAJOR SOURCES OF ENERGY

Energy sources are categorized into two main types:

2.1 Renewable Sources

Naturally replenished on a human timescale
Examples:

Solar energy – from the sun
Wind energy – from moving air
Hydropower – from moving water
Biomass – from organic matter
Geothermal energy – from Earth's internal heat
Tidal energy – from ocean tides

2.2 Non-Renewable Sources

Finite and depletable
Examples:

Fossil fuels – coal, petroleum, natural gas
Nuclear fuels – uranium and plutonium

3. ENERGY USE AND ITS IMPACT ON SOCIETY

Area	Impact
Social Development	Energy powers transportation, lighting, cooking, and communication. It supports education, healthcare, and water supply systems. Availability of energy enhances quality of life and economic opportunities.
Technological Advancement	Energy is central to the development of industries, manufacturing, and digital systems. Electrical energy supports AI, robotics, and internet technologies.
Health	Access to energy ensures clean water, storage of medicines, and operation of hospitals. However, pollution from fossil fuels can lead to respiratory diseases and cancer.
Environment	Fossil fuel use leads to air pollution, global warming, and climate change. Deforestation for biomass leads to habitat loss and reduced biodiversity. Oil spills, acid rain, and radioactive waste also harm the ecosystem.

4. ENERGY CONSERVATION AND SUSTAINABILITY

4.1 Definition

Energy conservation means using energy more efficiently and reducing wastage.

4.2 Methods of Energy Conservation

Using energy-saving appliances (LEDs, efficient refrigerators)
Switching off unused devices
Insulating buildings to reduce heating/cooling needs
Adopting public transport and reducing reliance on personal cars

4.3 Importance of Sustainable Energy Use

Reduces environmental degradation
Ensures availability of energy for future generations
Promotes economic resilience and reduces energy costs
Aligns with global climate goals like the **Paris Agreement**

5. EVALUATION OF DIFFERENT ENERGY SOURCES

Energy Source	Advantages	Limitations

Energy Source	Advantages	Limitations
Solar	Clean, renewable, low maintenance	Weather dependent, high setup cost
Hydro	Renewable, reliable, efficient	Ecosystem disruption, expensive
Wind	Renewable, zero emissions	Noise, aesthetic impact, variable output
Fossil Fuels	High energy output, easy transport	Pollution, finite, global warming
Nuclear	Large energy yield, low emissions	Radioactive waste, risk of accidents
Biomass	Reduces waste, renewable	Air pollution, deforestation risks

6. GLOBAL AND NATIONAL ENERGY CHALLENGES

Challenge	Description
Overdependence on Fossil Fuels	Leads to depletion and pollution

Challenge	Description
Energy Insecurity	Many regions lack access to reliable electricity (especially in rural Africa)
Climate Change	Resulting from greenhouse gas emissions tied to energy use
Geopolitical Conflicts	Energy resources can trigger tensions and wars
High Cost of Renewable Transition	Initial capital for solar, wind, and hydro projects is high

7. PRACTICAL SOLUTIONS TO ENERGY CHALLENGES

Solution	Description
Investment in Renewable Energy	Government and private sector support for solar, wind, and hydro projects
Public Awareness	Education campaigns on energy efficiency

Solution	Description
Policy Reforms	Enforce energy standards and encourage green innovation
Technology Transfer	Adopt clean technologies from developed nations
Diversification	Mix of energy sources to ensure reliability and reduce pressure on any single source

8. EVALUATION (WAEC/NECO PAST QUESTIONS)

Question 1 (WAEC 2019 Q4)

State two advantages and two disadvantages of nuclear energy.

Question 2 (NECO 2020 Q5)

Define energy and list four different forms of energy.

Question 3 (WAEC 2017 Q2)

Distinguish between renewable and non-renewable sources of energy with two examples each.

Question 4 (WAEC 2018 Q8)

Mention three ways by which energy can be conserved in homes and industries.

9. ASSIGNMENT

- Discuss three ways in which energy use affects the environment.
- With suitable examples, compare the advantages and disadvantages of hydroelectric power and fossil fuels.
- Suggest two realistic ways Nigeria can reduce its reliance on fossil fuels.
- Identify five everyday practices that promote energy conservation.